

University of Cagliari

Quantitative Macroeconomics

Problem Set 1

Group

N. F.
T. L.
F. M.

Professor

Prof. A. M.

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Introduction

This report studies two simple models from the first lectures of the course. In Exercise 1 we build a credit market with agents who live for two periods, solve for the equilibrium interest rate and use the model to read the rise in bond yields described in the article by Rampini. In Exercise 2 we use the static model of sectoral composition of Lecture 3 to compare South Korea and Turkey at the same income level, and then let income follow the data to see how much of their structural change the model can explain. All results and figures can be reproduced with the MATLAB scripts sent with the report.

1 Exercise 1: The Market for Credit and the Equilibrium Interest Rate

(a) The model: two agents and one market

Today the agent consumes or saves, $c_1 + s = w_1$. Tomorrow, instead, the agent can consume the income of $t = 2$ plus the savings increased by the interest: $c_2 = w_2 + (1 + r)s$. We obtain the budget constraint in present terms,

$$c_1 + \frac{c_2}{1+r} = w_1 + \frac{w_2}{1+r}. \tag{1}$$

The agent maximises $\ln c_1 + \beta \ln c_2$ subject to (1), from which we obtain the Euler equation $c_2 = \beta(1 + r)c_1$, which can also be written as the ratio between the two levels of consumption:

$$\boxed{\frac{c_2}{c_1} = \beta\,(1+r)} \tag{2}$$

The ratio of the marginal utilities is the marginal rate of substitution, $\text{MRS} = c_2/(\beta c_1) = 1 + r$, where $1+r$ is the relative price of consumption today in terms of consumption tomorrow: giving up one unit today gives $1+r$ units tomorrow. At the optimum the agent is indifferent to moving consumption from one period to the other.

We substitute $c_2 = \beta(1 + r)c_1$ in the budget constraint in present terms, obtaining the optimal levels of consumption in the two periods:

$$c_1^* = \frac{1}{1+\beta}\left(w_1 + \frac{w_2}{1+r}\right), \qquad c_2^* = \beta(1+r)\,c_1^* = \frac{\beta(1+r)}{1+\beta}\left(w_1 + \frac{w_2}{1+r}\right). \tag{3}$$

Consumption today is a fixed fraction $1/(1 + \beta)$ of permanent income: the higher β, the more patient the agent and the less she consumes today.

We have $s(r) = w_1 - c_1^*$. Defining $a \equiv \beta w_1/(1 + \beta)$ and $b \equiv w_2/(1 + \beta)$ we get

$$\boxed{s(r) = a - \frac{b}{1+r}} \tag{4}$$

The term a measures the push of the agents to lend (it grows with β and w_1) and b the push to borrow (it grows with w_2). The formula is valid for $1 + r > 0$.

The rate at which $s = 0$ is the autarky rate,

$$1 + \hat{r} = \frac{w_2}{\beta\, w_1} = \frac{b}{a}, \tag{5}$$

so the agent lends if $r > \hat{r}$ and borrows if $r < \hat{r}$. This is the same condition defined by the Euler equation (2): the agent lends if $\beta(1 + r) > w_2/w_1$.

What makes an agent a lender or a borrower. The basic idea is that the agent wants a consumption path that grows at the rate $\beta(1+r)$, while her income follows a given path that grows at the rate w_2/w_1. If the desired consumption grows more than income, the agent must carry resources from today to tomorrow: she lends. If it grows less, she must bring forward resources from tomorrow to today: she borrows. The three characteristics enter in this way:

- *Patience β.* A patient agent gives a lot of value to consumption tomorrow and wants consumption to grow over time, so he postpones consumption and tends to lend. An impatient agent wants to consume immediately: even with constant income ($w_2 = w_1$) he borrows, unless the interest rate is high enough to compensate him, that is $\beta(1 + r) > 1$.
- *Income today w_1.* The agent wants to spread her wealth over both periods, not to spend it all today. A high w_1 leaves resources to carry to the future, so the agent lends (saves). If w_1 is low, it is not enough to consume what she wants today and she borrows.
- *Income tomorrow w_2.* A high w_2 increases permanent income, and the agent wants to consume more today, with the aim, once again, of spreading resources over time. However, his income w_1 today has not changed, so he can do it only by borrowing against future income.

Further remarks

1. *The income profile matters, not the level.* The role of the agents depends on the ratio w_2/w_1: if both w_1 and w_2 double, consumption and saving double, but the sign of s does not change. The levels decide the size of saving, not its sign. An agent whose income falls over time (low w_2/w_1) tends to lend, one whose income grows (high w_2/w_1) tends to borrow.

2. *The role of the agent is not fixed: it depends also on r.* The same agent lends if the rate is high (postponing consumption pays a lot) and borrows if it is low. The autarky rate is the point of indifference.

3. *The equilibrium:* the market rate must lie above the autarky rate of the lender and below that of the borrower.

In equilibrium the funds lent are only the savings of the lenders, and all debts are repaid. The credit market is in equilibrium at the rate r^* at which the supply of credit equals the demand for credit, that is the sum of the savings is zero, $s_1(r^*) + s_2(r^*) = 0$. Substituting the saving function,

$$\boxed{1 + r^* = \frac{b_1 + b_2}{a_1 + a_2}} \tag{6}$$

With $\hat{r}_1 < r^* < \hat{r}_2$, agent 1 lends because r^* is above her autarky rate, and agent 2 borrows because r^* is below his. The agent with the lower autarky rate is the lender, and if the two rates were equal there would be no trade.

Table 1 reports the characteristics of the two agents, with the terms a and b of the saving function and the autarky rates $\hat{r}$.

	β	w_1	w_2	a	b	$\hat{r}$
Agent 1	0.95	15	5	7.3077	2.5641	-64.9%
Agent 2	0.90	10	20	4.7368	10.5263	122.2%

Table 1: Characteristics of the two agents, terms a and b of the saving function and autarky rates.

The first agent has an income that falls ($w_2/w_1 = 1/3$) and is patient, so she lends. The second has an income that doubles ($w_2/w_1 = 2$) and is more impatient, so he borrows. From the formula, $1 + r^* = 13.0904/12.0445 = 1.0868$, that is $r^* = 8.68\%$, between the two autarky rates. The quantity of credit exchanged is 4.948. Consumption is $c_1 = 10.05$, $c_2 = 10.38$ for agent 1 and $c_1 = 14.95$, $c_2 = 14.62$ for agent 2. Figure 1 shows the supply of credit (increasing in r) and the demand for credit (decreasing in r): the equilibrium is the point where they cross. The script also checks that $\sum_i s_i = 0$.

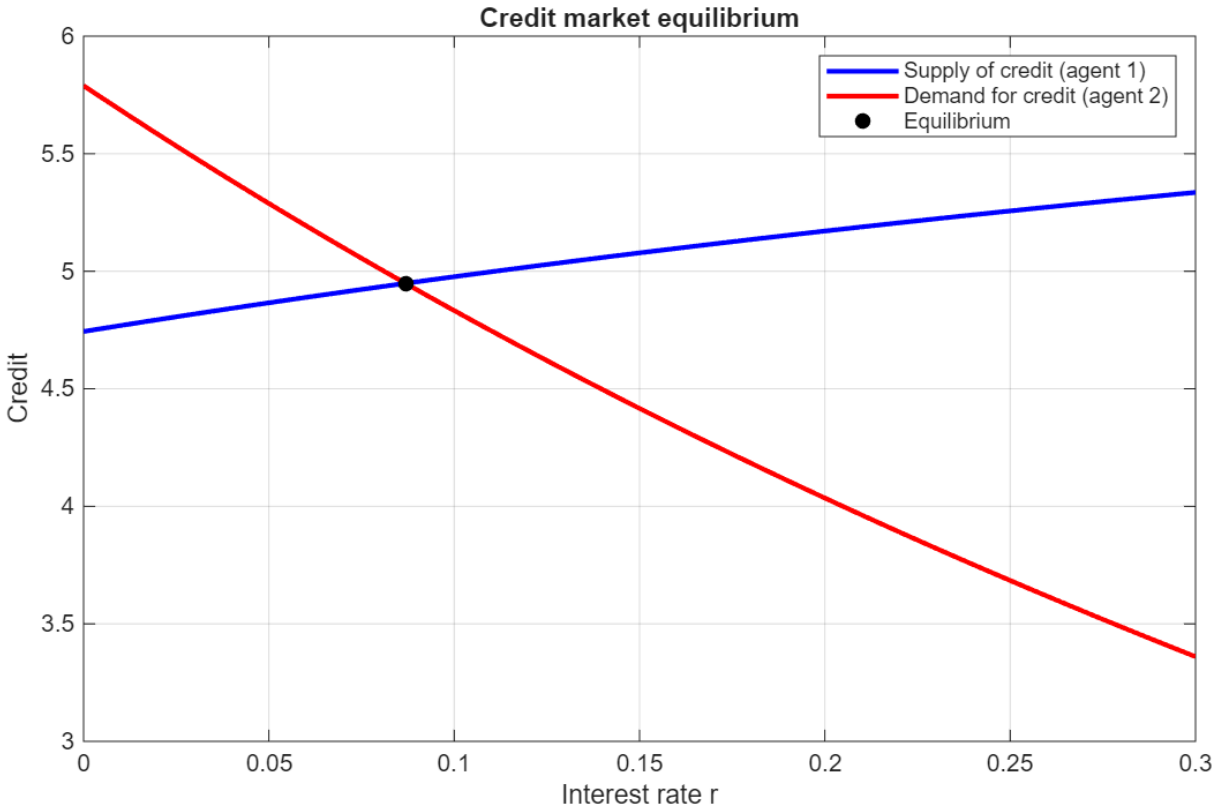


Figure 1: Supply and demand for credit as functions of r.

The figure shows, for $r \in [0, 30\%]$, the supply of credit $S(r) = s_1(r)$ (in blue), that is the saving of the lender, and the demand for credit $D(r) = -s_2(r)$ (in red), that is the borrowing of agent 2. The supply is increasing in r: a higher rate rewards more who postpones consumption, so agent 1 saves more. The demand is decreasing: a higher rate makes it more expensive to anticipate consumption, so agent 2 borrows less. The two curves cross at $r^* \approx 8.7\%$, where the credit exchanged is about 4.95. At a higher rate the supply exceeds the demand: lenders cannot find borrowers and the rate falls. At a lower rate the demand exceeds the supply: borrowers cannot find credit and the rate rises. The autarky rates, at which the supply and the demand are zero, are both outside the interval shown.

(b) Counterfactuals: what moves the interest rate?

We start from the economy of part (a) and change one characteristic at a time. A change modifies the saving function of only one agent, so it shifts one of the two curves, the supply $S(r) = s_1(r)$ or the demand $D(r) = -s_2(r)$. To read the explanations, recall the equilibrium rate of part (a),

$$1 + r^* = \frac{b_1 + b_2}{a_1 + a_2}, \qquad a_i = \frac{\beta_i \, w_{1,i}}{1 + \beta_i}, \quad b_i = \frac{w_{2,i}}{1 + \beta_i}, \tag{7}$$

and the quantity of credit, $s_1(r^*) = a_1 - \frac{b_1}{1+r^*} = -s_2(r^*)$.

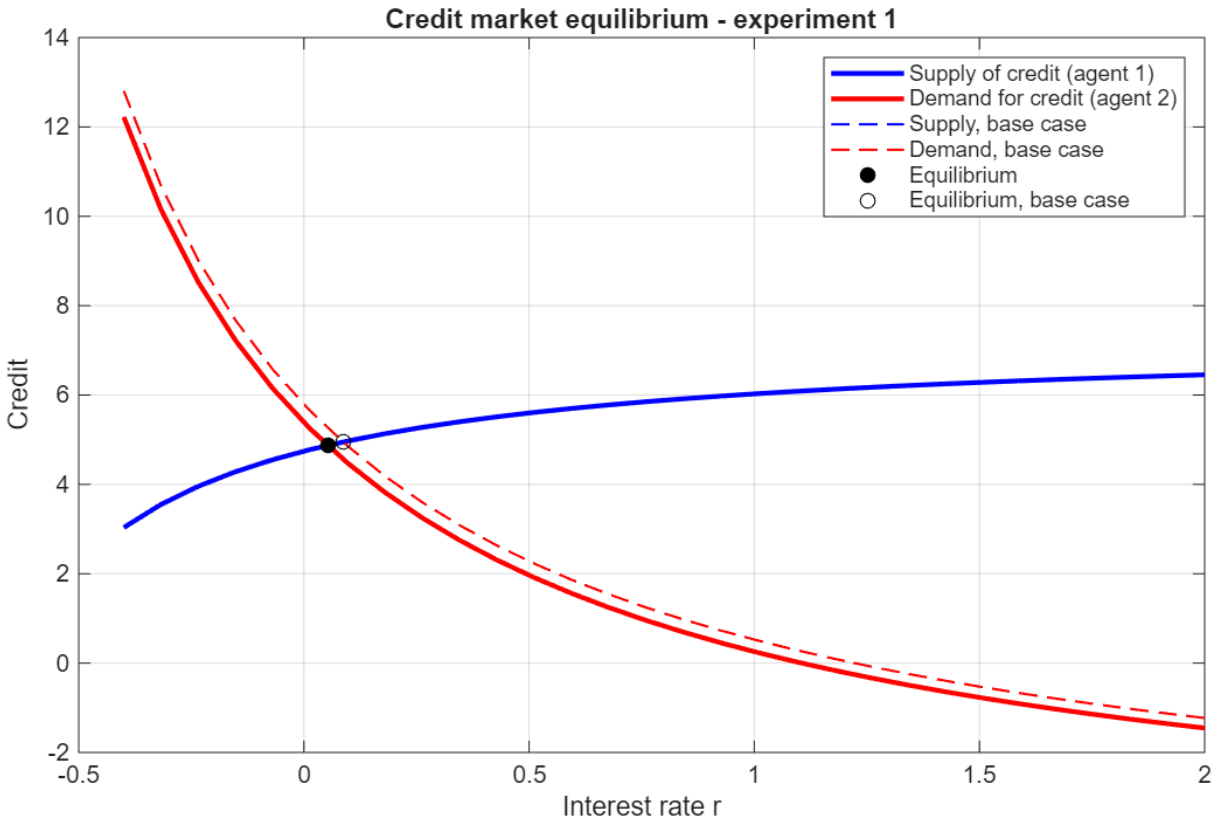


Figure 2: Experiment 1

Experiment 1: the borrower becomes more patient (β_2 from 0.90 to 0.95).

Prediction. A more patient borrower wants consumption to grow more over time, so he consumes less today and borrows less: the demand shifts down. We expect r^* to fall and credit to decrease, because at a lower rate the lender saves less.

Result: r^* goes from 8.68% to 5.26% and credit from 4.948 to 4.872.

In $s_2(r) = a_2 - \frac{b_2}{1+r}$ the term a_2 rises and b_2 falls, so the borrower saves more (borrows less) at every rate. In the formula of r^* the numerator falls and the denominator rises. Credit decreases only a little because the shift is small and the lender reacts little to the rate.

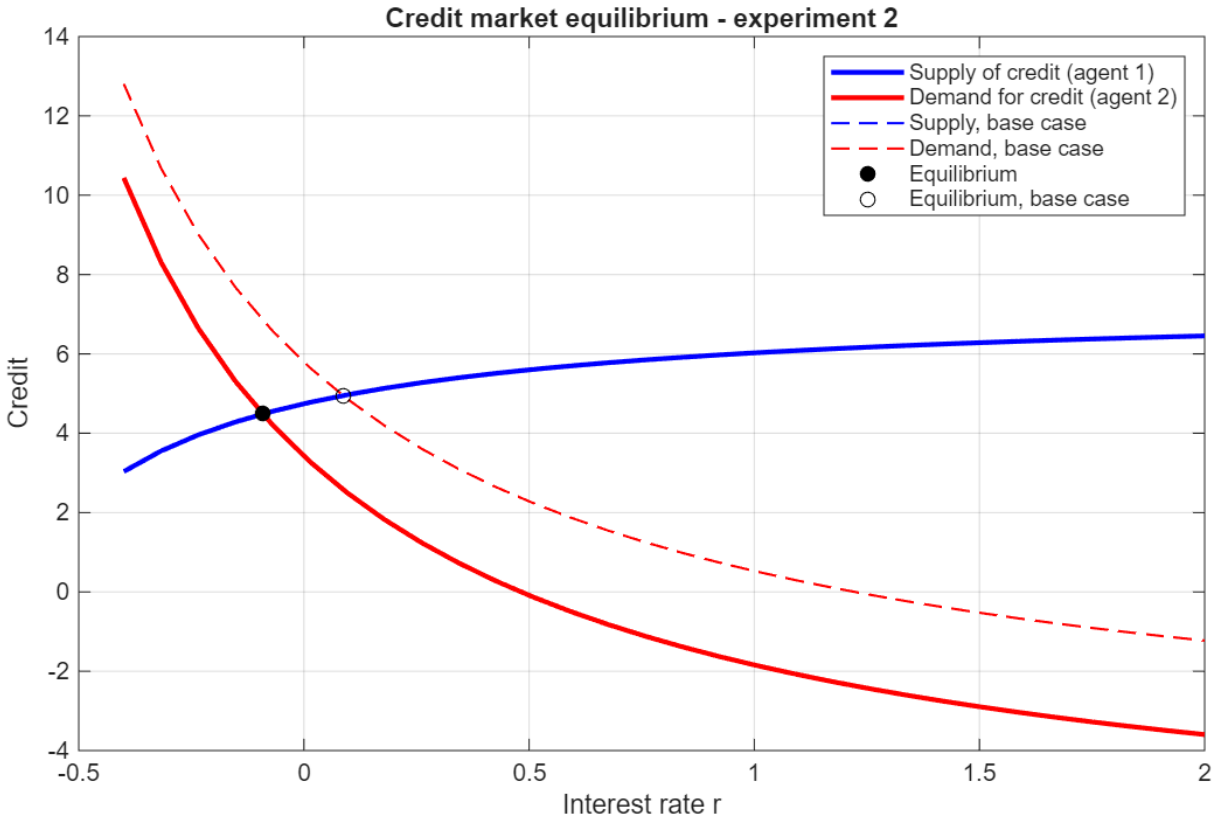


Figure 3: Experiment 2

Experiment 2: the borrower has a higher income today ($w_{1,2}$ from 10 to 15).

Prediction. The borrower is richer today and needs less to anticipate consumption: he borrows less and the demand shifts down. We expect r^* and credit to fall.

Result: $r^* = -9.18\%$ and credit 4.484.

The term a_2 rises a lot while b_2 does not change, so s_2 rises and the demand falls strongly. The autarky rate of the borrower falls from 122% to 48%, but r^* stays below it, so he keeps borrowing. The rate is negative because the lender, who is patient and has an income that falls over time, accepts to lend even at a negative return: she lends as long as $r > \hat{r}_1 = -64.9\%$.

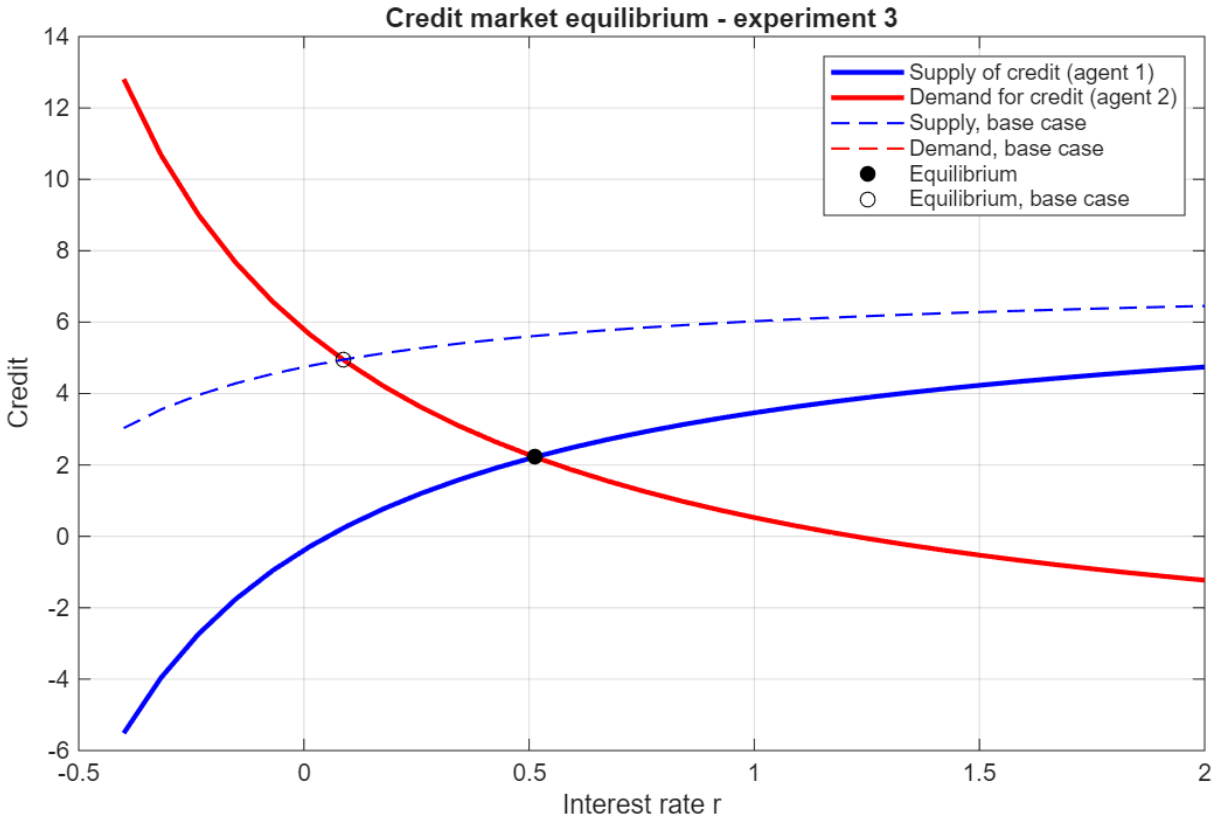


Figure 4: Experiment 3

Experiment 3: the lender has a higher income tomorrow ($w_{2,1}$ from 5 to 15).

Prediction. The lender knows she will be richer tomorrow and wants to consume more today: she saves less and the supply shifts down. To convince her to lend, a higher rate is needed, so we expect r^* to rise. We expect credit to fall, because at a higher rate the borrower borrows less.

Result: $r^* = 51.26\%$ and credit 2.222.

The term b_1 rises, so $s_1(r) = a_1 - \frac{b_1}{1+r}$ falls at every rate: at $r = 0$ the lender would even borrow. Her autarky rate rises from -64.9% to 5.3%, and she lends only above it. This is the strongest effect of the three experiments.

Summary of the results

	r^* (%)	Quantity of credit
Base case	8.68	4.948
1. Exp.1	5.26	4.872
2. Exp.2	-9.18	4.484
3. Exp.3	51.26	2.222

Table 2: Equilibrium interest rate and quantity of credit in the base case and in the three experiments.

(c) Many agents

With N_1 agents of type 1 and N_2 agents of type 2, each with the saving function $s_i(r) = a_i - \frac{b_i}{1+r}$ of part (a), the market is in equilibrium when the sum of the savings is zero:

$$N_1\, s_1(r^*) + N_2\, s_2(r^*) = 0 \implies 1 + r^* = \frac{N_1 b_1 + N_2 b_2}{N_1 a_1 + N_2 a_2}. \tag{8}$$

If half of the agents are of each type, $N_1 = N_2 = N_{tot}/2$, the factor $N_{tot}/2$ cancels out between numerator and denominator and we go back to the formula of part (a), equation (6).

First point: $N_{tot} = 500$ **and** $N_{tot} = 1000$**.**

- *What does not change:* $r^* = 8.68\%$. The saving of each agent, $s_i(r^*) = a_i - \frac{b_i}{1+r^*}$, depends only on r^* and on the characteristics of the agent, so consumption and the credit per lender (4.948) and per agent (2.474) are the same as in part (a).
- *What changes:* the total credit, $\frac{N_{tot}}{2}\, s_1(r^*)$, which is proportional to N_{tot}: 1237.1 with 500 agents and 2474.2 with 1000 (against 4.948 with two agents).

- *Why:* N_{tot} multiplies both the supply and the demand of credit, so the crossing of the two curves moves in quantity but not in price (r). The size of the economy changes the size of the market, not its price.

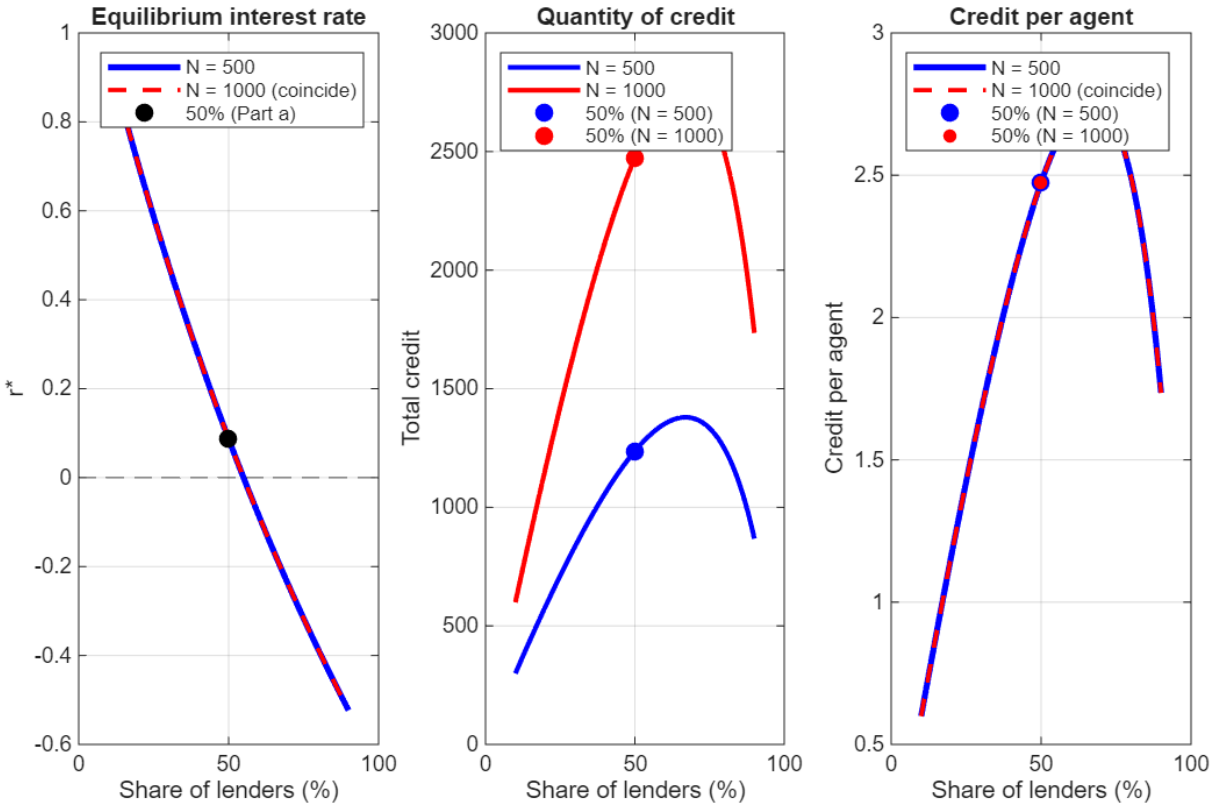


Figure 5: Equilibrium interest rate and total credit as functions of the share of lenders.

Second point: how to obtain a different r^* with $N_{tot} = 1000$.

We keep $N_{tot} = 1000$ and the characteristics of the two types of agents fixed, and we change the composition of the population, that is the share of lenders. From the equilibrium condition (8), r^* can move only if the ratio N_1/N_2 changes.

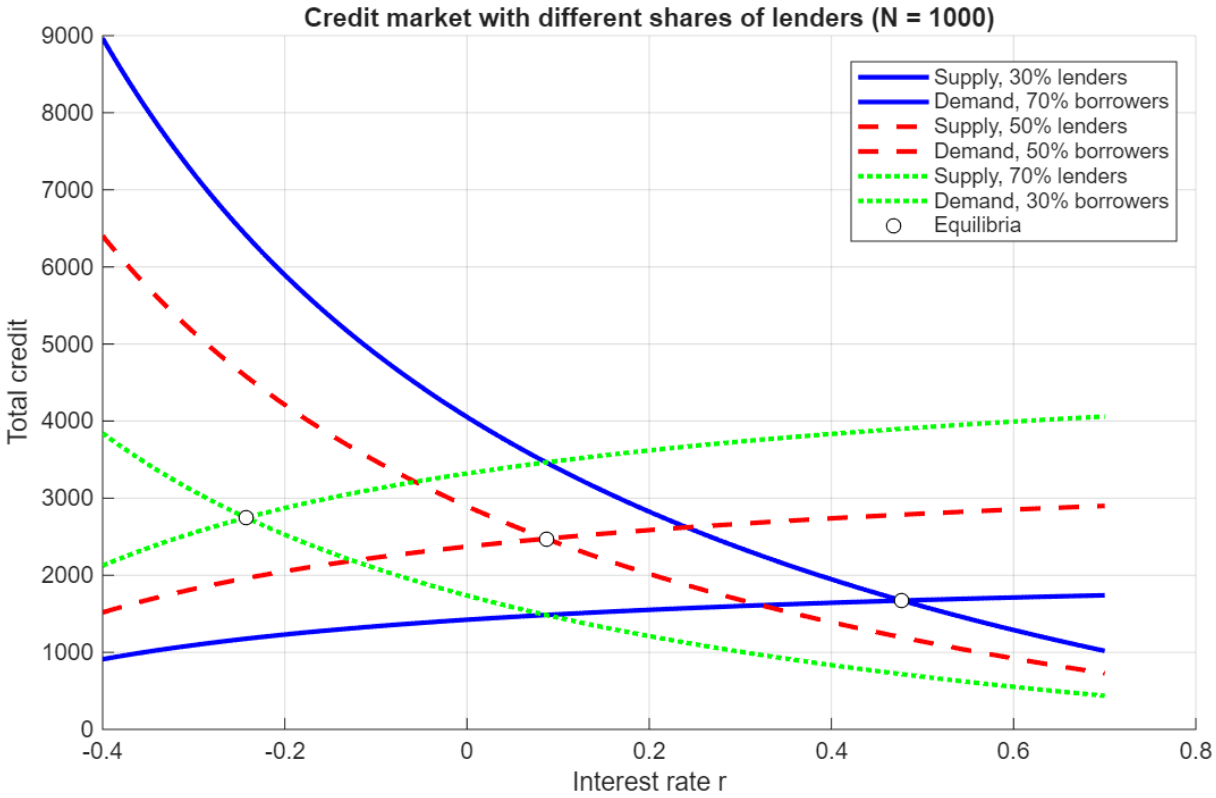


Figure 6: Supply and demand for credit with different shares of lenders ($N = 1000$).

To see why the rate moves, note that the supply of credit is $S(r) = N_1 s_1(r)$ and the demand is $D(r) = -N_2 s_2(r)$. With N_{tot} fixed, increasing the number of lenders moves agents from the side of the borrowers to the side of the lenders, so the supply rises and the demand falls at every rate. Figure 6 shows the supply and the demand of credit for three compositions. With 30% of lenders (blue solid lines) the supply is low and the demand is high, so the curves cross at a high rate: $r^* = 47.74\%$, with a credit of 1671.6. With 50% (red dashed lines) we are in the economy of part (a): $r^* = 8.68\%$ and credit 2474.2. With 70% (green dotted lines) the supply is high and the demand is low: $r^* = -24.23\%$ and credit 2746.6. As the share of lenders rises, the curves move and the crossing slides to lower interest rates.

Back to the article.

In our model a borrower is any agent who wants to spend more today than his income, and a lender is any agent who saves. The State, the firms and the savers of the article can be read in this way. Table 3 summarises the answers.

In the article	In our model	Experiment	r^*
A State that runs a larger deficit	More borrowers, or borrowers who borrow more	Part (c), lenders from 50% to 30%	↑
Firms that expect large returns from AI	A borrower with a higher income tomorrow, $w_{2,2}$	Not simulated, from equation (6)	↑
A country with many domestic savers	More lenders	Part (c), lenders from 50% to 70%	↓
Good news	A higher income tomorrow, w_2	Experiment 3	↑

Table 3: The article in the language of the model.

A State that runs a larger deficit. A deficit means that the State spends more than its revenues today and borrows the difference: in the model it is a borrower, and a larger deficit adds demand for credit. The closest experiment is the second point of part (c): with 30% of lenders instead of 50%, r^* rises from 8.68% to 47.74%. In that experiment the total number of agents is fixed, so more borrowers also means fewer lenders, and total credit falls (from 2474.2 to 1671.6). A State that borrows more without reducing the number of savers would only shift the demand up, so r^* would rise and credit would rise too.

Firms that invest in artificial intelligence. Our agents do not invest, but a firm that expects large returns tomorrow behaves like a borrower with a higher income tomorrow, $w_{2,2}$: it wants to spend today against its future income. We did not run this experiment, but the formula of r^* gives the answer. The term $b_2 = w_{2,2}/(1+\beta_2)$ is in the numerator of equation (6), so r^* rises. With the values of Table 1 and $w_{2,2}$ from 20 to 25, the formula gives $r^* = 30.5\%$. The demand shifts up, so credit rises as well, to 5.34.

A country with a large pool of domestic savers. Savers are lenders. With 70% of lenders the supply of credit is high and the demand is low, and r^* falls to -24.23% (part (c)). Patience works in the same direction: in Experiment 1 a more patient borrower saves more and r^* falls from 8.68% to 5.26%.

Can good news raise interest rates? Yes. Good news about the future means that agents expect a higher income tomorrow. Everybody wants to bring part of this income forward and consume more today, so fewer resources are saved and savings become scarce. In the model a higher w_2 raises b, the numerator of r^*. Experiment 3 is exactly this case: the lender expects a higher income tomorrow and r^* rises from 8.68% to 51.26%. A high interest rate is therefore not always a bad sign: it can mean that agents are optimistic about the future.

The article ends by asking which interest rate is needed to convince someone to finance all this. In the model the answer is r^*: the rate must rise until the savings that lenders want to offer are equal to what borrowers want to borrow.

2 Exercise 2: One Model, Two Countries

Data and choice of the countries

We compare South Korea, a high-income country, with Turkey, an upper-middle-income one (World Bank classification). The pair suits the exercise well: Turkey today is about as rich as Korea was in 1997, so the sample contains the same stage of development in both countries, reached at very different speeds. Korea grew fast and regularly, Turkey more slowly. For the sectors we use employment shares. All the series come from the World Development Indicators (Table 4).

Variable	Unit
GDP per capita	constant 2015 US$
Employment in agriculture	% of total employment
Employment in manufacturing	% of total employment
Employment in services	% of total employment

Table 4: World Development Indicators (World Bank), 1992–2025.

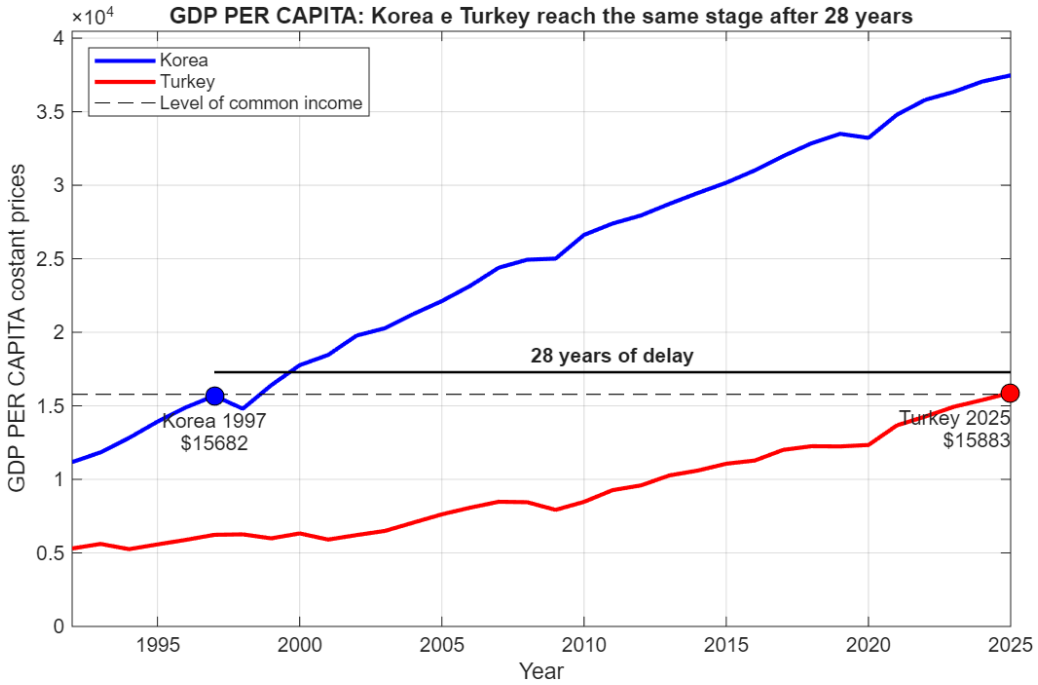


Figure 7: GDP per capita at constant prices, Korea and Turkey.

(a) Static comparison

Korea in 1997 and Turkey in 2025 have almost the same GDP per capita, 15,682 and 15,883 dollars (a gap of 1.3%), even though 28 years separate them (Figure 7). This is the comparable stage we use. With fixed prices, income is the only thing that moves the shares in the model, so two countries with the same income and the same preferences would have the same sectoral composition. The data are close to this for services, but not for the other two sectors: Turkey has 4 more points in agriculture, Korea 5 more in manufacturing (Table 5).

	GDP per capita (US$)	Agriculture	Manufacturing	Services
Korea 1997	15,682	10.2	32.4	57.3
Turkey 2025	15,883	14.2	27.3	58.5

Table 5: The two comparable stages in the data (shares in %).

We use the household of Lecture 3, which chooses c_a, c_m and c_s to maximise

$$C = \left[\omega_a^{1/\varepsilon} \left(c_a + \bar{c}_a\right)^{\frac{\varepsilon-1}{\varepsilon}} + \omega_m^{1/\varepsilon} \left(c_m + \bar{c}_m\right)^{\frac{\varepsilon-1}{\varepsilon}} + \omega_s^{1/\varepsilon} \left(c_s + \bar{c}_s\right)^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}} \tag{9}$$

subject to

$$p_a c_a + p_m c_m + p_s c_s = w, \tag{10}$$

with $\omega_a + \omega_m + \omega_s = 1$. A negative $\bar{c}_a$ works as a minimum quantity of food, so agriculture is a necessity; a positive $\bar{c}_s$ is like services produced at home, which makes services a luxury. The first order conditions are

$$C^{\frac{1}{\varepsilon}} \, \omega_i^{\frac{1}{\varepsilon}} \left(c_i + \bar{c}_i\right)^{-\frac{1}{\varepsilon}} = \lambda p_i, \qquad i = a, m, s. \tag{11}$$

Together with the budget constraint, this is the system that `solve_consumption_foc.p` solves numerically. Solving it by hand is still useful, because it shows what drives the shares. Adding $\sum_j p_j \bar{c}_j$ to both sides of the budget constraint gives the corrected income $\tilde{w}$,

$$\sum_j p_j \left(c_j + \bar{c}_j\right) = w + \sum_j p_j \bar{c}_j \equiv \tilde{w}, \tag{12}$$

and we get the expenditure share of sector i:

$$s_i = \frac{p_i c_i}{w} = \frac{\omega_i \, p_i^{1-\varepsilon}}{\omega_a p_a^{1-\varepsilon} + \omega_m p_m^{1-\varepsilon} + \omega_s p_s^{1-\varepsilon}} \, \frac{\tilde{w}}{w} \; - \; \frac{p_i \bar{c}_i}{w}. \tag{13}$$

With all prices equal to one the denominator is 1 and $\tilde{w} = w + \bar{c}_a + \bar{c}_m + \bar{c}_s$, so

$$s_i = \omega_i \, \frac{w + \bar{c}_a + \bar{c}_m + \bar{c}_s}{w} - \frac{\bar{c}_i}{w} = \omega_i + \frac{\omega_i \left(\bar{c}_a + \bar{c}_m + \bar{c}_s\right) - \bar{c}_i}{w}. \tag{14}$$

Here ε has disappeared: with equal prices it has no effect on the shares.

Equation (14) also tells us how to choose the weights. Inverting it,

$$\omega_i = \frac{s_i \, w + \bar{c}_i}{w + \bar{c}_a + \bar{c}_m + \bar{c}_s}, \tag{15}$$

and the three weights add up to one by themselves, since the numerators sum to $w + \bar{c}_a + \bar{c}_m + \bar{c}_s$. The other parameters are the same in both countries (Table 6), so the two economies differ only in the weights.

Parameter	Value	Choice
$\bar{c}_a$, $\bar{c}_s$	-1388, 12251 (both countries)	Subsistence level for agriculture, services as a luxury.
$\bar{c}_m$	0	As in the lecture.
ε	0.75	As in the lecture (complements). No effect when prices are equal.
p_a, p_m, p_s	1	Normalisation: no data on sectoral prices.
ω_a, ω_m, ω_s	by country	From the formula above, to match the observed shares exactly.

Table 6: Parameter choices in the static model.

The hint of the exercise suggests looking at $\bar{c}_a$ and $\bar{c}_s$ relative to income. In our case this does not separate the two countries: incomes are almost equal, so the two terms weigh about the same in both (Table 7), and the whole difference has to go into the weights.

	ω_a	ω_m	ω_s	w	$\bar{c}_a/w$	$\bar{c}_s/w$
Korea 1997	0.0082	0.1917	0.8001	15682	-8.9%	78.1%
Turkey 2025	0.0325	0.1621	0.8054	15883	-8.7%	77.1%

Table 7: Calibrated weights and value of $\bar{c}_a$, $\bar{c}_s$ relative to income.

With these weights the model gives exactly the observed shares (Table 8 and Figure 8). So the calibration can reproduce both countries at the comparable stage, but this holds by construction (two free weights for two independent shares), and the real test of the model comes in part (b).

	Agriculture	Manufacturing	Services
Korea 1997, model	10.24	32.45	57.32
Korea 1997, data	10.24	32.45	57.32
Turkey 2025, model	14.22	27.29	58.49
Turkey 2025, data	14.22	27.29	58.49

Table 8: Sectoral shares (%) at the comparable stage of development.

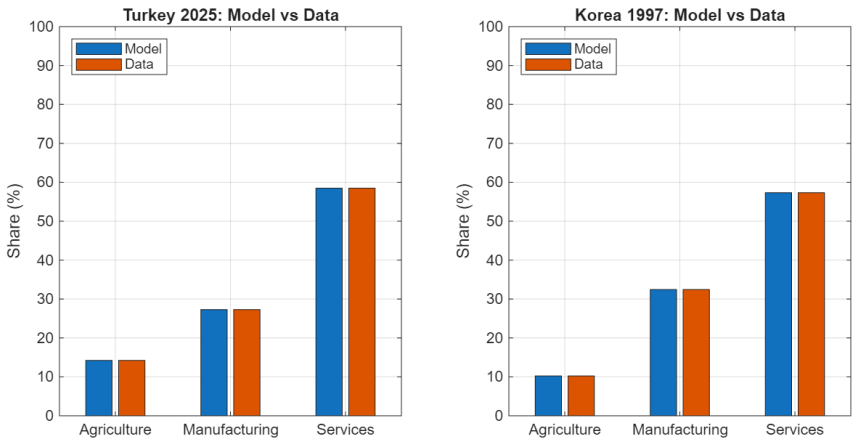


Figure 8: Sectoral shares at the comparable stage of development: model and data.

Turkey's weight on agriculture is four times the Korean one (0.033 against 0.008), and its weight on manufacturing is lower (0.162 against 0.192). Taken literally, Turkish households simply like food more and manufactured goods less. We do not think this is the right reading. The weights are the only free parameters, so they absorb every force that the model leaves out, and part of what looks like a difference in tastes is really an error in how the model describes the two economies.

Two of these forces seem important to us. The first is productivity. If Turkish agriculture is less productive, it needs more workers to produce the same food, and food is relatively more expensive. With $\varepsilon < 1$ a higher p_a raises the agricultural share (equation (13)), but our prices are fixed at one, so the model can only show this as a larger ω_a. The second is manufacturing. At the same income Korea is more industrialised than Turkey (32.4% of employment in manufacturing against 27.3%): its growth was driven by exports of manufactured goods, so many Korean workers produce goods that are sold abroad. The household of the model only consumes, so the model can only show this as a larger ω_m.

(b) Dynamic comparison

The household does not change in the dynamic version. The model has no capital and no saving, so every year it solves the static problem of part (a) with a new income, which we set equal to GDP per capita in the data:

$$w_t = \text{GDP per capita}_t, \qquad t = 1992, \ldots, 2025. \tag{16}$$

All the other parameters keep their values from part (a), and we solve the model year by year with `fsolve` (the figures start in 1997). For Turkey this means that we target the final shares (2025), not the initial ones: 2025 is the comparable stage of part (a), so the same parameters serve both parts. The cost is that the error moves to the start of the period, which we discuss below. With prices equal to one, each share depends only on income:

$$s_{i,t} = \omega_i + \frac{\omega_i\left(\bar{c}_a + \bar{c}_m + \bar{c}_s\right) - \bar{c}_i}{w_t}, \qquad \frac{\partial s_{i,t}}{\partial w_t} = -\frac{\omega_i\left(\bar{c}_a + \bar{c}_m + \bar{c}_s\right) - \bar{c}_i}{w_t^2}. \tag{17}$$

The sign of the numerator gives the direction. It is positive for agriculture, so that share falls as income grows, and negative for services, which rise. Manufacturing is the case to keep in mind: $\bar{c}_m = 0$ and the sum of the $\bar{c}_i$ is positive (10,863), so its numerator is positive and the manufacturing share can only fall. The model cannot produce a rising or hump-shaped manufacturing share.

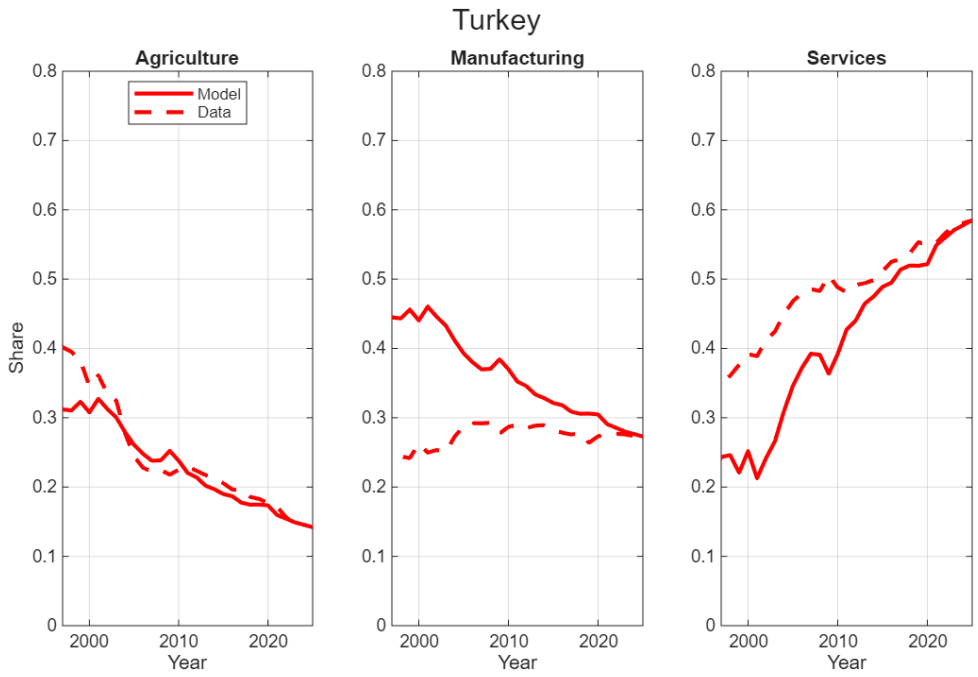


Figure 9: Structural transformation in Turkey: sectoral shares, model (solid) and data (dashed).

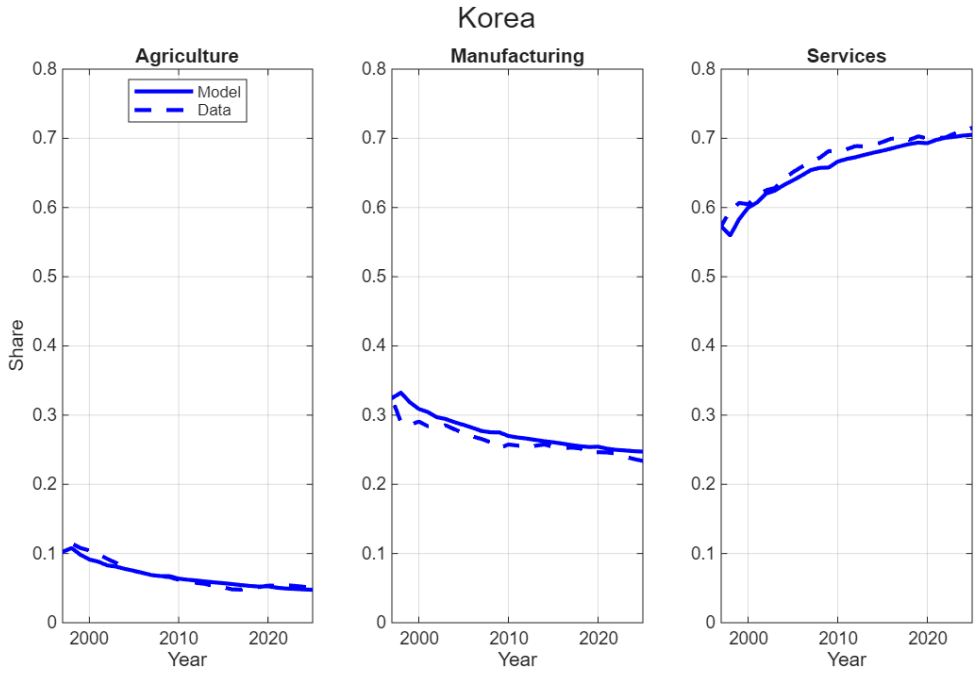


Figure 10: Structural transformation in Korea: sectoral shares, model (solid) and data (dashed).

		1997		2025	
		Model	Data	Model	Data
Korea	Agriculture	10.2	10.2	4.8	5.1
	Manufacturing	32.4	32.4	24.7	23.4
	Services	57.3	57.3	70.5	71.5
Turkey	Agriculture	31.2	40.2	14.2	14.2
	Manufacturing	44.5	24.8	27.3	27.3
	Services	24.3	35.0	58.5	58.5

Table 9: Model and data at the start and at the end of the period (shares in %).

For Korea the model works well (Figure 10 and Table 9). Some of this is expected, since the $\bar{c}_i$ were set on Korea, but it still tells us that most of the Korean structural change after 1997 is what income growth alone would predict.

Turkey is the more interesting case. 2025 is matched by construction, and the further we go back in time, the worse the fit becomes (Figure 9). The largest error is in manufacturing. In the data its share is flat or slightly rising, from 24.8% to 27.3%, while in the model it can only fall: to end at 27.3% in 2025 it has to start at 44.5% in 1997, far above the real value. Services go wrong the other way. In 1997 Turkey had about 6,230 dollars per person, so $\bar{c}_s/w \approx 2$ and the household in the model is too poor to buy many services (24% of the total), while in the data 35% of workers were already in services. Agriculture is closer, with the right direction but 9 points too low at the start. In all three sectors the model gives Turkey the same income mechanism that fits Korea, and Turkey's history does not follow it.

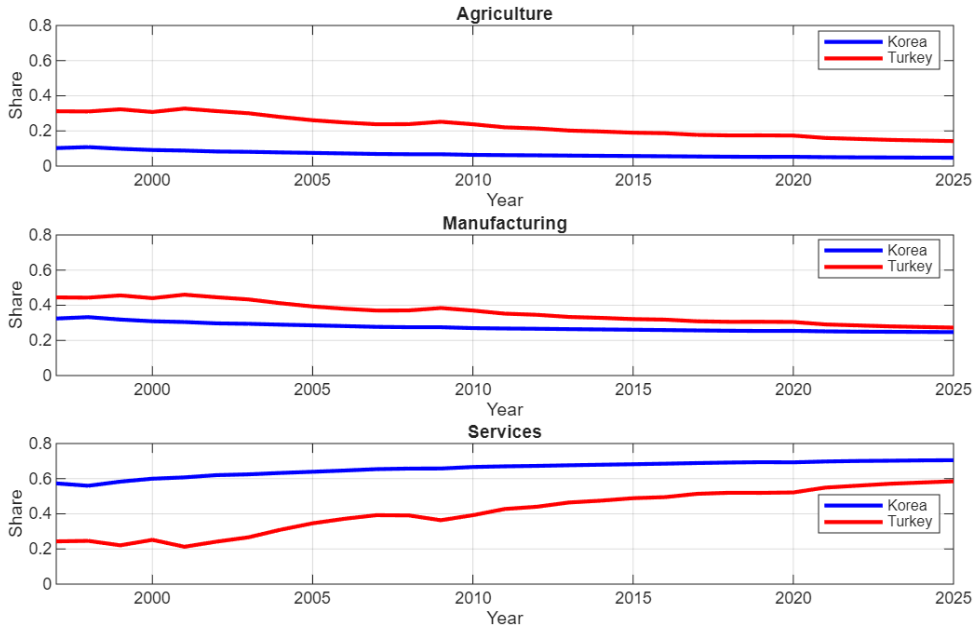


Figure 11: Sectoral shares in the model: Korea (blue) and Turkey (red).

Figure 11 puts the two model economies side by side and helps with the last question, why the same transformation took a different number of years. In the model the shares depend only on income (equation (17)), so the time of a transformation is simply the time needed to grow from one income level to the other. Table 10 takes the same interval, from about 11,200 to 15,800 dollars, in both countries. Korea covered it in 5 years, growing 7.0% a year; Turkey needed 9, at 3.9%, and with more crises along the way (GDP per capita fell in 1994, 1999, 2001 and 2009, in Korea only in 1998 and 2020).

	Period	GDP p.c. (US$)	Years	Growth	Agriculture	Services
Korea	1992–1997	11,177 → 15,682	5	7.0%	13.3 → 10.2	49.7 → 57.3
Turkey	2016–2025	11,280 → 15,883	9	3.9%	19.7 → 14.2	52.5 → 58.5

Table 10: The same income interval in the two countries. Growth is the average per year; shares are in the data, in %.

The table also shows something the model does not predict. Over the same income interval Turkey moved more workers out of agriculture than Korea (5.5 points against 3.1), while in the model the same change in income gives a similar change in the shares. And the model cannot say why growth was so different in the first place: income is an input of the model, not a result.

A Derivations for Exercise 1

Substituting $s = w_1 - c_1$ in the consumption of the second period,

$$c_2 = w_2 + (1+r)(w_1 - c_1) \implies (1+r)\,c_1 + c_2 = (1+r)\,w_1 + w_2. \tag{18}$$

This is the constraint in terms of future goods, that is measured in units of tomorrow's consumption. Dividing by $1+r$ we obtain the budget constraint in present terms,

$$c_1 + \frac{c_2}{1+r} = w_1 + \frac{w_2}{1+r}. \tag{19}$$

The two equations are equivalent, but from the latter it is more immediate to see that the present value of total consumption must equal permanent income, that is the present value of the income of the two periods.

The agent maximises $\ln c_1 + \beta \ln c_2$ subject to (1), with Lagrangian

$$\mathcal{L} = \ln c_1 + \beta \ln c_2 + \lambda \left[w_1 + \frac{w_2}{1+r} - c_1 - \frac{c_2}{1+r} \right]. \tag{20}$$

We obtain the first order conditions by differentiating the Lagrangian with respect to c_1 and c_2:

$$\frac{\partial \mathcal{L}}{\partial c_1} = \frac{1}{c_1} - \lambda = 0 \implies \lambda = \frac{1}{c_1}, \qquad \frac{\partial \mathcal{L}}{\partial c_2} = \frac{\beta}{c_2} - \frac{\lambda}{1+r} = 0. \tag{21}$$

Substituting $\lambda = 1/c_1$ in the second condition, $\beta/c_2 = 1/((1+r)c_1)$, from which we obtain the Euler equation $c_2 = \beta(1+r)c_1$, which can also be written as the ratio between the two levels of consumption:

$$\boxed{\frac{c_2}{c_1} = \beta\,(1+r)} \tag{22}$$

We substitute $c_2 = \beta(1+r)c_1$ in the budget constraint in present terms, obtaining the optimal levels of consumption in the two periods:

$$c_1 + \frac{\beta(1+r)\,c_1}{1+r} = w_1 + \frac{w_2}{1+r} \implies (1+\beta)\,c_1 = w_1 + \frac{w_2}{1+r}. \tag{23}$$

Therefore

$$c_1^* = \frac{1}{1+\beta}\left(w_1 + \frac{w_2}{1+r}\right), \qquad c_2^* = \beta(1+r)\,c_1^* = \frac{\beta(1+r)}{1+\beta}\left(w_1 + \frac{w_2}{1+r}\right). \tag{24}$$

We have $s(r) = w_1 - c_1^*$. We put everything on the same denominator and substitute c_1^*:

$$s = \frac{(1+\beta)\,w_1 - w_1 - \dfrac{w_2}{1+r}}{1+\beta} = \frac{\beta\,w_1 - \dfrac{w_2}{1+r}}{1+\beta}. \tag{25}$$

Separating the two terms and defining $a \equiv \beta w_1/(1+\beta)$ and $b \equiv w_2/(1+\beta)$ we get

$$\boxed{s(r) = a - \frac{b}{1+r}} \tag{26}$$

The denominator of the saving function, $1+\beta$, is always positive ($\beta > 0$), so the sign of s is the sign of the numerator. An agent lends if $s > 0$, that is $\beta w_1 > w_2/(1+r)$. We multiply by $1+r > 0$ and divide by $\beta w_1 > 0$:

$$\boxed{1+r > \frac{w_2}{\beta\,w_1}} \quad \text{(lender)}, \qquad \boxed{1+r < \frac{w_2}{\beta\,w_1}} \quad \text{(borrower)}. \tag{27}$$

In equilibrium the funds lent are only the savings of the lenders, and all debts are repaid. The credit market is in equilibrium at the rate r^* at which the supply of credit equals the demand for credit, that is the sum of the savings is zero, $s_1(r^*) + s_2(r^*) = 0$. Substituting the saving function,

$$a_1 - \frac{b_1}{1+r^*} + a_2 - \frac{b_2}{1+r^*} = 0 \implies a_1 + a_2 = \frac{b_1+b_2}{1+r^*} \implies \boxed{1+r^* = \frac{b_1+b_2}{a_1+a_2}} \tag{28}$$

Writing a_i and b_i explicitly,

$$1+r^* = \frac{\dfrac{w_{2,1}}{1+\beta_1} + \dfrac{w_{2,2}}{1+\beta_2}}{\dfrac{\beta_1 w_{1,1}}{1+\beta_1} + \dfrac{\beta_2 w_{1,2}}{1+\beta_2}}. \tag{29}$$

B Derivations for Exercise 2

The Lagrangian is

$$\begin{aligned}\mathcal{L} = &\left[\omega_a^{1/\varepsilon}(c_a+\bar{c}_a)^{\frac{\varepsilon-1}{\varepsilon}} + \omega_m^{1/\varepsilon}(c_m+\bar{c}_m)^{\frac{\varepsilon-1}{\varepsilon}} + \omega_s^{1/\varepsilon}(c_s+\bar{c}_s)^{\frac{\varepsilon-1}{\varepsilon}}\right]^{\frac{\varepsilon}{\varepsilon-1}} \\ &+ \lambda\left[w - p_a c_a - p_m c_m - p_s c_s\right]\end{aligned} \tag{30}$$

and the first order conditions are

$$C^{\frac{1}{\varepsilon}}\,\omega_i^{\frac{1}{\varepsilon}}\,(c_i+\bar{c}_i)^{-\frac{1}{\varepsilon}} = \lambda p_i, \qquad i = a,m,s. \tag{31}$$

If we divide the condition of good i by the one of services, C and λ cancel out:

$$\left(\frac{\omega_i}{\omega_s}\right)^{\frac{1}{\varepsilon}}\left(\frac{c_i+\bar{c}_i}{c_s+\bar{c}_s}\right)^{-\frac{1}{\varepsilon}} = \frac{p_i}{p_s} \quad\implies\quad c_i+\bar{c}_i = \frac{\omega_i}{\omega_s}\left(\frac{p_i}{p_s}\right)^{-\varepsilon}(c_s+\bar{c}_s). \tag{32}$$

Adding $\sum_j p_j\bar{c}_j$ to both sides of the budget constraint gives the corrected income $\tilde{w}$,

$$\sum_j p_j\,(c_j+\bar{c}_j) = w + \sum_j p_j\bar{c}_j \equiv \tilde{w}, \tag{33}$$

and substituting the previous result we get the expenditure share of sector i:

$$s_i = \frac{p_i c_i}{w} = \frac{\omega_i\,p_i^{1-\varepsilon}}{\omega_a p_a^{1-\varepsilon} + \omega_m p_m^{1-\varepsilon} + \omega_s p_s^{1-\varepsilon}}\,\frac{\tilde{w}}{w} \;-\; \frac{p_i\bar{c}_i}{w}. \tag{34}$$

Equation (14) also tells us how to choose the weights. Inverting it,

$$s_i\,w + \bar{c}_i = \omega_i\,(w + \bar{c}_a + \bar{c}_m + \bar{c}_s) \quad\implies\quad \omega_i = \frac{s_i\,w + \bar{c}_i}{w + \bar{c}_a + \bar{c}_m + \bar{c}_s}, \tag{35}$$